

CAPSTONE PROJECT

DIGIWISE – AGENTIC AI FOR DIGITAL FINANCIAL LITERACY

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OUTLINE

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PROBLEM STATEMENT

As India moves toward a digital economy, millions of people are using UPI, QR codes, net banking, and digital wallets — but many lack awareness about how to use these tools safely.

This results in increased financial frauds, QR scams, and confusion around budgeting, credit scores, and digital payments, especially among rural or non-tech-savvy users. There is a need for a trustworthy, multilingual AI assistant that can explain digital finance in simple language and help users make safer financial decisions.

Proposed Solution:

DigiWise is an AI-powered multilingual assistant built using IBM watsonx.ai and the Granite LLM. It provides safe, simple, and context-aware guidance on digital finance topics like UPI, budgeting, fraud prevention, and bank accounts — especially for Indian users.

It acts as a smart Agentic AI that uses trusted documents, reasons over user queries, and responds clearly in English, Hindi, or Marathi.

TECHNOLOGY USED

- IBM watsonx.ai Prompt Lab – for building the AI assistant
- Granite-13b-instruct LLM – IBM's large language model for reasoning and response generation
- Retrieval Augmented Generation (RAG) – uses RBI & SEBI PDFs as trusted context
- Python (Jupyter Notebook) – for testing and execution
- IBM Cloud Runtime Environment – to run and manage the agent
- GitHub – for version control and public hosting

IBM CLOUD SERVICES USED

- IBM watsonx.ai Prompt Lab
For designing, testing, and running the AI assistant using prompt engineering.
- IBM Cloud Watsonx AI runtime
To execute the prompts and models securely in the cloud.
- IBM Agent Builder (Prompt Flow)
For configuring agent goals, rules, instructions, and behavior settings.
- IBM Granite foundation model
Large Language Model (LLM) used for natural, human-like responses.

WOW FACTORS

This AI assistant helps bridge the gap between digital technology and financial literacy by offering a safe, multilingual, and easy-to-use platform. DigiWise stands out with its agentic behavior - it not only responds but reasons over user goals using trusted documents like RBI and SEBI guides.

Unique features:

Agentic AI behavior: the assistant understands intent, applies reasoning, and follows a goal-oriented flow.

Multilingual query handling: users can ask questions in English, Hindi, or Marathi.

Safety-first design: avoids hallucinations, personal financial advice, and promotional suggestions.

API-ready code samples in Python, Java, JavaScript, Scala, and cURL for easy integration.

Explains complex financial terms such as UPI, EMI, credit score, and savings accounts in simple, user-friendly language.

END USERS

- **General public** using UPI, QR codes, or digital banking in daily life
- **First-time digital finance users** in rural or semi-urban areas
- **Senior citizens and non-English speakers** who need guidance in Hindi or Marathi
- **College students and young professionals** looking to understand budgeting, loans, and credit scores
- **NGOs and financial literacy campaigners** aiming to spread awareness
- **Educational institutions** conducting workshops on digital financial safety
- **Customer support bots or banks** wanting to integrate a safe, AI-based financial assistant

RESULTS



DigiWise 08:30 PM

Welcome to DigiWise

Hey! I'm DigiWise, your smart assistant for all things digital finance.



Sample questions

How can I use UPI safely for online payments?



What is a savings account and how do I open one?



RESULTS

The screenshot displays the IBM watsonx Agent Lab interface. The top navigation bar includes the IBM watsonx logo, an 'Upgrade' button, a help icon, a notification bell, and user account information for 'Kusum Choudhary's Account' in 'Dallas'. Below this, the breadcrumb trail shows 'Projects / DigiWise / Agent Lab'. The main interface is split into two panels. The left panel, titled 'Build', contains a 'Setup' section with fields for 'Name' (DigiWise) and 'Description' (Hey! I'm DigiWise, your smart assistant for all things digital finance.). It also features an 'Icon' section with a placeholder image and a 'Quick start questions' section with a placeholder text. The right panel, titled 'Agent preview', shows a chat conversation. The first message is from the user (KC) asking 'How can I use UPI safely to send money?'. The second message is from the agent (DigiWise) providing a detailed response with seven steps for using UPI safely. The chat input field at the bottom right contains the text 'Type something...'.

IBM watsonx Upgrade ? Bell Kusum Choudhary's Account Dallas KC

Projects / DigiWise / Agent Lab Share feedback Unsaved New agent + Deploy

Build AI Model: llama-3-3-70b-instruct

Setup

Name

DigiWise

Description 72/100

Hey! I'm DigiWise, your smart assistant for all things digital finance.

Icon Placeholder image

Quick start questions

Add up to 4 quick start questions that your users can select.

Agent preview

It is safe to link your bank account with Google Pay as Google Pay uses secure servers to store payment information and keeps your full card details secure. Additionally, Google Pay has security infrastructures in place to help keep your account safe. However, it's always important to be cautious when sharing personal and financial information online and to ensure that you're using a legitimate and secure connection when linking your bank account to Google Pay.

► How did I get this answer?

KC You: 07:46 PM

How can I use UPI safely to send money?

DigiWise 07:46 PM

You can use UPI safely to send money by following these steps:

1. Ensure you have a valid UPI ID or VPA (Virtual Payment Address) and a bank account linked to it.
2. Use a secure and trusted UPI app, such as Google Pay, PhonePe, or Paytm.
3. Enter the recipient's UPI ID or VPA correctly to avoid any errors.
4. Use a strong password and enable two-factor authentication to secure your transactions.
5. Keep your mobile device and UPI app updated with the latest security patches.
6. Be cautious of phishing scams and never share your UPI PIN or personal information with unknown sources.
7. Use a reliable and secure internet connection when making transactions.

Type something...

RESULTS

Projects / DigiWise / DigiWise

Share feedback 3 Autosaved 9:13 PM New agent + Deploy

Build

AI Model: granite-3-3-8b-instruct

Setup

Configuration

Framework LangGraph Architecture ReAct

Instructions

Advanced configuration

You are a helpful assistant that uses tools to answer questions in detail.
When greeted, say "Hi, I'm DigiWise, your AI finance assistant. How can I help you?"

You are smart, friendly AI assistant built using IBM watsonx.ai to help users understand digital finance in a safe and simple way. It acts as an Assistant AI, meaning it can reason, use...

Knowledge

Tools

Add a tool Create custom tool

Agent preview

KC You: 09:12 PM

मुझे यूपीआई फ्रॉन्ट से कैसे बचना चाहिए?

DigiWise 09:12 PM

यूपीआई फ्रॉन्ट से बचने के लिए, इन सुझावों का पालन करें:

1. विश्वसनीय ऐप्स का उपयोग करें: केवल आधिकारिक UPI ऐप्स जैसे BHIM, Google Pay, PhonePe, या Paytm का उपयोग करें। ये ऐप सुरक्षित हैं और व्यापक रूप से उपयोग किए जाते हैं।

2. प्राप्तकर्ता का सत्यापन करें: किसी भी लेनदेन को शुरू करने से पहले हमेशा प्राप्तकर्ता के UPI ID या VPA (वर्चुअल पेमेंट एड्रेस) की जाँच करें। सुनिश्चित करें कि यह इरादे के प्राप्तकर्ता के साथ मेल खाता है।

3. 2-फैक्टर ऑथेंटिकेशन (2FA) सक्षम करें: अपने UPI ऐप को PIN या बायोमेट्रिक प्रमाणीकरण (फिंगरप्रिंट या फेशियल रिकॉग्निशन) के साथ सुरक्षित करें, एक अतिरिक्त सुरक्षा परत के लिए।

4. सार्वजनिक वाई-फाई से बचें: सार्वजनिक वाई-फाई नेटवर्क पर UPI लेनदेन करने से बचें क्योंकि वे कम सुरक्षित हो सकते हैं। एक विश्वसनीय नेटवर्क या मोबाइल डेटा का उपयोग करें।

5. सॉफ्टवेयर अपडेट रखें: अपने UPI ऐप और मोबाइल ऑपरेटिंग सिस्टम को नियमित रूप से अपडेट करें ताकि आप नवीनतम सुरक्षा पैच का लाभ उठा सकें।

6. लेनदेन की निगरानी करें: अनधिकृत गतिविधि के लिए अपने बैंक स्टेटमेंट और UPI लेनदेन इतिहास की नियमित रूप से जाँच करें।

edunet
foundation

RESULTS

Deployed AI Agent

DigiWise ✓ Deployed Online

API reference

Test

Preview

New chat +

Welcome to DigiWise

Hey! I'm DigiWise, your smart assistant for all things digital finance.



Quick start samples

How can I use UPI safely for online payments?

What is a savings account and how do I open one?

Type something...



About this deployment ×

Name

DigiWise

Description

Hey! I'm DigiWise, your smart assistant for all things digital finance.

Deployment Details

Deployment ID: 8b26e7f7-fe90-45...

Serving name:

No serving name.

Software specification:

[runtime-24.1-py3.11](#) ⚙️

Hardware specification:

Extra extra small: 1 CPU and 2 GB RAM

Copies:

1

Tags

wx-agent

Associated asset

[DigiWise](#)

5c4e1583-4506-4cb7-a712-95b874fc20a7

CONCLUSION

- DigiWise demonstrates how AI can be purposefully applied to address real-world challenges like digital financial literacy and scam prevention.
- It is designed with ethical AI principles, avoiding hallucinations and personal financial advice, while empowering users through simplified explanations.
- By combining IBM watsonx.ai, Granite LLM, and Retrieval-Augmented Generation (RAG), this assistant delivers context-aware, multilingual, and safe responses — making financial knowledge more accessible and reliable.

FUTURE SCOPE

- Voice-based interaction for rural or less literate users
- WhatsApp or mobile app integration
- Scam alert notifications
- Secure integration with banking apps or customer support bots
- Dynamic financial tutorials
- Region-specific customization

IBM CERTIFICATIONS



Getting Started with AI

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Kusum Choudhary

for the completion of

**Lab: Retrieval Augmented Generation with
LangChain**

(ALM-COURSE_3824998)

According to the Adobe Learning Manager system of record

Completion date: 24 Jul 2025 (GMT)

Learning hours: 20 mins

Rag lab Certificate

GITHUB LINK

- Github Link - https://github.com/Kusumchoudhary08/DigiWise_Bot



THANK YOU